

MOHAMED SALEM

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✉ [Email](#) — [LinkedIn](#) — [GitHub](#) — [Codeforces](#)

Summary

Java Backend Developer skilled in Java SE/EE and the Spring ecosystem. Experienced in building scalable applications with Spring Boot, RESTful APIs, and JWT authentication. Proficient in JPA/Hibernate for persistence, multithreading, and Java 8+ (Streams, Lambdas, Optional). Strong in IoC (DI, AOP), Docker containerization, and Redis caching. Familiar with Maven, Swagger, and JavaFX. Solid understanding of concurrency, OS concepts, and S.O.L.I.D principles. Currently focusing on Microservices, cloud-native systems, and distributed design with Kafka, Keycloak, and Stripe integrations.

Education

Bachelor of Computer Science

2022 - 2026

Menofia University, Egypt

GPA: 3.3 / 4.0

Projects

Poly-Clinic-System (Microservices Architecture)

[GitHub Repository](#)

- Built a production-grade distributed microservices ecosystem using Spring Boot, PostgreSQL, Kafka, Redis, MapStruct, Mail Sender, Cron Job Scheduler, Stripe, Docker, and Keycloak.
- Used Redis distributed locks to prevent race conditions across concurrent financial and booking operations.
- Implemented Cron-based schedulers to automate service maintenance routines and optimize memory usage.
- Implemented synchronous REST + Feign communication with automatic JWT injection and asynchronous Kafka events using the Outbox Pattern for exactly-once delivery.
- Developed enterprise IAM with Keycloak, API Gateway security filters, centralized RBAC, and a Token Service for uniform authentication across services.
- Integrated Stripe manual-capture payments with idempotency, and fault-tolerant Kafka-based notifications.

E-Commerce Backend API (Spring Boot)

[GitHub Repository](#)

- Developed a RESTful API using Java and Spring Boot with clean layered architecture and Hibernate ORM for persistence with complex entity relationships.
- Implemented JWT authentication with refresh token flow, secure access/refresh token rotation, and Redis-based token blacklist for logout and revocation.
- Built product, category, cart, and order management modules using JPA/Hibernate with advanced mappings (OneToMany, ManyToMany, cascading, and orphan removal).
- Implemented a Global Exception Handler with custom exceptions for consistent error responses across the system.

Gitlet (Distributed Version Control System – Java)

[GitHub Repository](#)

- Developed a lightweight version control system in Java with Git-like commands (init, add, commit, checkout, branch, merge).
- Used snapshotting and SHA-1 hashing for version tracking with persistent storage.
- Designed data structures (commits, blobs, branches) and DAG traversal for merge base detection.
- Gained experience with file I/O, serialization, and CLI development.

Pathfinding Visualizer (JavaFX – Algorithms Simulation)

[GitHub Repository](#)

- Built visualization tool demonstrating BFS, DFS, and Dijkstra algorithms on dynamically generated weighted grid.
- Implemented user interactions for setting start, goal, and obstacles with real-time algorithm progression.
- Used JavaFX Tasks and multithreading to ensure smooth UI rendering without blocking the main application thread.
- Applied clean OOP design and efficient data structures to optimize traversal and visualization performance.

Achievements

- Ranked **44th** on Day 2 of the ECPC 2024 Qualifications among 400+ teams.
- Reached Codeforces **Pupil** rank — [Leo.Salem](#).
- Solved **2000+** problems on competitive programming platforms.

Technical Skills

Languages: Java (SE, EE, FX), C, C++, Python, JavaScript

Frameworks: Spring Boot, Spring MVC, Spring Cloud (Eureka, Gateway, Feign), Hibernate (JPA), Servlets, JSP, JavaFX

Tools: Git, Maven, Postman, Grafana, Zipkin, Swagger UI, Docker, Kubernetes

AI Tools: Claude, ChatGPT, IntelliJ AI Assistant, Cursor, GitHub Copilot

Databases: MySQL, PostgreSQL, JDBC, JPQL, Redis

Concepts: REST APIs, Microservices, JWT, RBAC, IoC, Event-Driven Architecture (Kafka, RabbitMQ)

Foundations: OOP, Multithreading, SOLID, Concurrency, OS, Data Structures, Algorithms